Design of Local Area DGNSS Architecture to Support Unmanned Aerial Vehicle Networks: Concept of Operations and Safety Requirements Validation

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ABSTRACT

A local-area differential global navigation satellite systems (LAD-GNSS) supports unmanned aerial vehicles (UAVs) with high integrity and accuracy. This study investigates three major issues in fully establishing LAD-GNSS and analyzes their performance. First, we define the concept and requirements of UAV operation, including segregation of UAV operation coverage airspaces, and the derivation of navigation requirements, such as alert limits (ALs), for each operational coverage. Second, we design an LAD-GNSS architecture by simplifying the hardware and monitoring algorithms of both the ground facility and onboard module, using the well-established ground-based augmentation system (GBAS) as a starting point. Lastly, we perform theoretical performance evaluations comparing position uncertainty bounds, which are represented by protection levels (PLs), to the corresponding navigation requirements for each coverage. We derive and compare PLs and ALs under both nominal and malfunction cases. In addition, we describe a method for deriving PLs for excessive acceleration and code-carrier divergence fault scenarios, which are bounded by using a maximum-allowable error in range. Using these PLs, integrity/continuity allocations are ideally and dynamically assigned to each single-fault hypothesis to obtain optimized PLs that are identical for all fault scenarios. We find that vertical protection levels (VPLs) are reduced by approximately 16% when implementing the optimal allocation method.

1. INTRODUCTION

One of the most essential technologies required for operating unmanned aerial vehicles (UAVs) is reliable and accurate computation of navigation solutions. Local-area UAV networks that utilize local-area differential global navigation satellite system (LAD-GNSS) solutions have been proposed to achieve high navigation accuracy and reliability, which are required for autonomous UAV applications [1]. LAD-GNSS meets integrity requirements comparable to those of ground-based augmentation system (GBAS) Category I–III operations by monitoring navigation faults at both the ground station and UAV and by broadcasting integrity information to the UAV. By utilizing integrity information, which defines navigation system error models, UAVs can compute their conservative position error bounds (i.e., protection levels (PLs)) and consequently maintain safe separation distances in real time. The previously proposed LAD-GNSS architecture for a local-area UAV network includes several methods of simplifying the current GBAS integrity monitoring algorithms and hardware configurations to lower the system's cost and complexity while maintaining acceptable safety [1]. The performance and integrity of LAD-GNSS can also be improved by designing additional onboard modules. A LAD-GNSS integrity monitor testbed was installed on the KAIST campus. However, these modifications, including ground hardware and software changes and airborne components, have not yet been fully analyzed and implemented from a system-design perspective.

In this study, we develop several key components of the LAD-GNSS architecture, including system operation coverage and requirements, modifications of integrity monitors in both ground and airborne subsystems, and theoretical performance evaluations. The operation requirements—vertical alert limits (VALs)—are derived from the operating environment and navigation uncertainty. Herein, we compare the vertical protection level (VPL) under nominal and malfunction cases to the VAL to determine if the system meets the integrity requirements for a given operation.

The LAD-GNSS integrity monitors were developed using GBAS as a starting point. In this study, we implement major modifications of the ground subsystems by designing additional onboard modules. First, instead of maintaining the ionospheric geometry screening at the ground subsystem, which is computationally expensive owing to the computation of ionospheric error for the all-in-view geometry, we utilize simplified airborne geometry screening. In this onboard module, an error induced by an ionospheric anomaly is computed for a single geometry that the UAV uses for its positioning. This can lower conservatism and computation burden significantly.

In the current GBAS architecture, separate PL values for three scenarios—fault-free (H0), single reference-receiver fault (H1), and ephemeris-fault cases—are defined based on the predetermined integrity and continuity allocations. These allocations can ideally be dynamically assigned to make the PLs identical for all fault scenarios. The use of UAV information provides a significant advantage over the conventional GBAS and enables real-time allocation. Since the onboard module has information about the satellite geometry utilized by the UAV for its positioning, it is possible to compute the PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

The remainder of this paper is organized as follows. Section 2 illustrates the LAD-GNSS architecture, including the operation coverage and integrity monitoring modifications. Section 3 explains the operation requirement of LAD-GNSS, and Section 4 details the methods used in our theoretical performance evaluation. Section 5 discusses our conclusions.

2. LAD-GNSS ARCHITECTURE FOR UAV

The proposed LAD-GNSS architecture for UAVs emphasizes low costs and complexity for commercial UAV applications with compact ground/onboard installation units, while supporting high navigation accuracy and integrity within the coverage airspace.

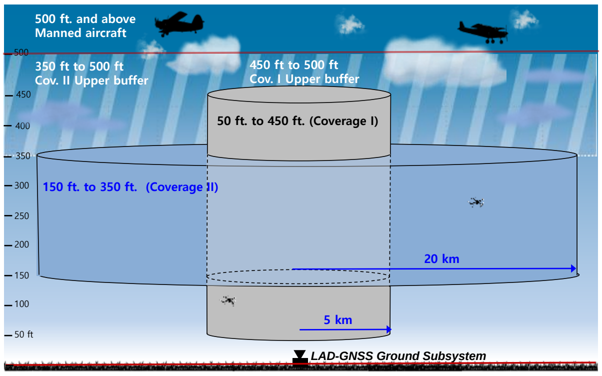

Figure 1. LAD-GNSS architecture for UAVs (assuming the absence of ground obstacles)

Figure 1 illustrates the operational concept of LAD-GNSS architecture for UAVs when there are no ground obstacles within operation regions. The ground facility shown at the center is the source of the LAD-GNSS corrections and integrity information, as well as real-time guidance for each UAV operating within the coverage airspace. To enable safe UAV operations in civil airspace, coverage airspaces are defined around the ground facility. The radii of the proposed system's coverage airspaces are defined based on expected UAV missions for each region. Coverage I supports UAV operations within a radius of 5 km from the ground facility. Coverage I is designed to potentially support the UAV landing phase down to its lower limit with high-accuracy and high-integrity navigation. Coverage II is designed to support high-speed operations, with lower navigation accuracy than Coverage I, within a radius from 5 to 20 km from the ground facility. The range of operation altitudes for each coverage airspace is determined by considering correction accuracy over the distance from the operating UAVs to the ground facility.

Supported altitude for Coverage I is defined from 50 to 450 ft; the upper and lower margins for vertical position uncertainty from both upper (500 ft) and lower (0 ft) hard bounds are 50 ft. The supported altitude range for Coverage II is defined from 150 ft to 350 ft by assuming margins of 150 ft on both the upper and lower sides. Note that the upper and lower margins for Coverage II are larger than those for Coverage I to account for the degraded correction accuracy of a UAV operating at a farther distance.

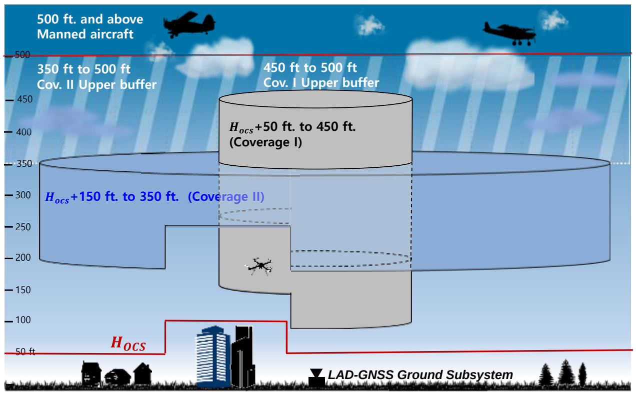

Figure 2. LAD-GNSS architecture for UAV (when H_{OCS} are assumed)

Table 1. Coverage airspace of LAD-GNSS

Criteria	Radius (from the ground facility)	Altitude
Coverage I	~ 5km	50 ft+H_{OCS} ~ 450 ft
Coverage II	5km~20km	150 ft +H_{OCS} ~350 ft

The operational concept introduced in Fig. 1 can be applied in ideal conditions, in which there are no ground obstacles around operational regions. However, this assumption is unrealistic for various commercial UAV operations. Thus, we introduce the concept of height of obstacle clearance surface (H_{OCS}), which indicates the maximum height of an obstacle that can be placed along the aircraft approach path. H_{OCS} have already been defined for GBAS, and are enforced at runways that have precision approaches, non-precision approaches, and even visual operations [2]. Since the environments that UAVs fly above are more variable than those for manned aircraft, H_{OCS} for UAVs should be adaptively defined based on these environments. The H_{OCS} can be defined prior to operation based on environmental conditions; the operational concept introduced in Fig. 1 is simply a special case in which H_{OCS} is 0. Figure 2 shows the operation concept with a pre-modeled H_{OCS}. H_{OCS} is modeled to bound the heights of buildings and trees, which should not be violated for safety reasons. Assuming that H_{OCS} is not 0, the operational altitude ranges are variable and reduced. The defined coverage airspaces are summarized in Table 1.

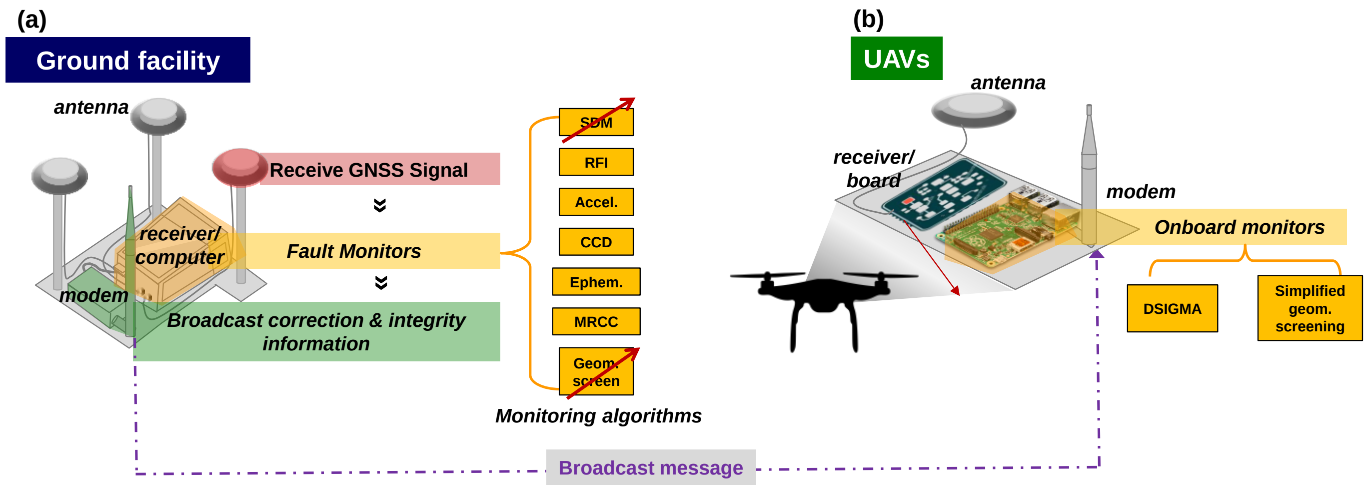

Figure 3. Configuration and components of the LAD-GNSS architecture

Figure 3 describes the simplified configuration and components of LAD-GNSS, which include the ground facility (Fig. 3(a)) and UAV onboard module (Fig. 3(b)). Three ground reference receivers are connected to low-cost antennas. Considering that siting constraints for most commercial applications will prevent having room to spread ground antennas over hundreds of meters, antennas are sited close together. Low-quality antennas and relatively short antenna separation increase the impact of multipath errors. To mitigate performance degradation due to this hardware restriction, and to take advantage of the new system configuration, this study develops several key components of the LAD-GNSS architecture, including optimal integrity allocations and integrity monitor modifications. Optimal integrity allocations allocate the total system integrity risk budget dynamically to make the final position uncertainty bounds identical for all fault modes, which reduces the final user position uncertainty bound compared to the original bound computed from pre-allocated integrity budgets. We introduce the algorithm for optimal integrity allocation in Section 4.1. In addition to optimal allocation of the integrity budget, monitor modifications can potentially improve overall performance. To mitigate the integrity risk caused by anomalous ionosphere gradients, a dual smoothing ionosphere gradient monitor algorithm (DSIGMA) (which is adopted for GBAS CAT-II/III) is also employed, with a code-carrier divergence (CCD) monitor to meet system requirements. In addition, we move geometry screening from the ground facility to onboard to reduce conservatism and the computational burdens based on known user satellite geometry onboard. This simplified geometry screening computes position uncertainty under ionospheric anomaly conditions in real time for a single satellite geometry tracked by the user, instead of computing impacts for all possible user geometries. We discuss the modified geometry screening and position uncertainty simulations in Section 4.2. Lastly, the signal deformation monitor (SDM) can be eliminated by selecting ground and airborne GNSS receivers that are either the same or very similar, which cancels out the error resulting from users applying differential corrections [1].

3. OPERATION REQUIREMENT FOR LAD-GNSS FOR UAVs

We compare PLs to ALs in each position axis of interest to determine if the system meets integrity and continuity requirements for a given operation. In this study, we analyze the VALs of Coverage I and II based on the total system error (TSE), specifically its composition of the sum of the navigation sensor error (NSE) and flight technical error (FTE). It is sufficient to consider only vertical requirements, since they are more difficult to fulfill than the lateral ones.

The key requirement for vertical TSE performance of the LAD-GNSS is ensuring that actual UAV location does not violate the H_{ocs} or manned aircraft zones. For example, the vertical TSE of a UAV operating in Coverage I must not exceed 50 ft, because the lower limit of Coverage I is defined as $50\text{ ft} + H_{ocs}$. Corresponding to these requirements, we divide our analysis into a nominal NSE case and a malfunction NSE case. In the nominal case, no faults occur (i.e., a fault-free condition) or faults are detected by monitors in the navigation system (faulted condition); in the malfunction case, an unmonitored fault that causes a large position error occurs.

3.1 VALs under Nominal NSE Case

In the analysis of the nominal NSE case, the allowable probability of exceeding the limit for the fault-free condition is 10^{-6}; the allowable probability of exceeding the limit for the faulted condition is 10^{-5}. These allowed risk can be changed depends on system design or application. This study handles only the fault-free condition for the VAL of nominal NSE case ($VAL_{nominal}$) because it has a tighter requirement than that of the faulted condition. To facilitate our analysis, we use simplified assumptions in our analytical derivation. We assume that the relationship between the vertical NSE and $VAL_{nominal}$ at the worst geometry is expressed as

$$\sigma_{NSE,vert}(m) = \frac{VAL_{nominal}(m)}{k_{FFMD}}; \quad k_{FFMD} = 5.43, \tag{1}$$

where k_{FFMD} is a multiplier defined to ensure a specified probability of fault-free missed detection for cases with three ground reference receivers. The vertical TSE is normally distributed with a zero mean and standard deviation ($\sigma_{TSE,vert}$) given by

$$\sigma_{TSE,vert} = \sqrt{\sigma^2_{NSE,vert} + \sigma^2_{FTE,vert}}. \quad (2)$$

We assume that, if the TSE is zero, the UAV height is the lower limit (50 ft from a H_{OCS} for Coverage I or 150 ft from a H_{OCS} for Coverage II). Therefore, the probabilities that the UAV height is lower than the H_{OCS} are given by

$$P_{risk,Coverage\ I} = \int_{50\ ft}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,vert}} e^{-\frac{x^2}{2\sigma^2_{TSE,vert}}} dx, \quad (3)$$
$$P_{risk,Coverage\ II} = \int_{150\ ft}^{\infty} \frac{1}{\sqrt{2\pi}\sigma_{TSE,vert}} e^{-\frac{x^2}{2\sigma^2_{TSE,vert}}} dx.$$

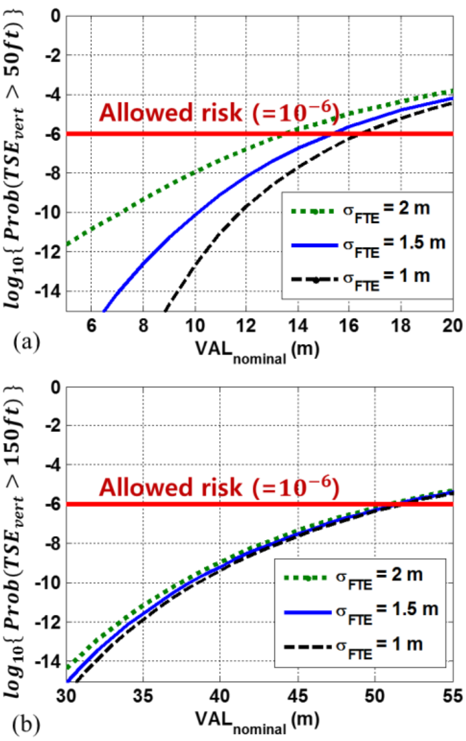

Figure 4. Probability that TSE_{vert} is greater than (a) 50 ft for Coverage I and (b) 150 ft for Coverage II

Figure 4(a) and 4(b) show plots of the fault-free risk versus $VAL_{nominal}$ when $\sigma_{FTE,vert} = 1\ m, \sigma_{FTE,vert} = 1.5\ m$, and $\sigma_{FTE,vert} = 2m$. Previous work has shown that $\sigma_{FTE,vert}$ of the multi-copter UAV is less than a meter [3]. Based on the conservative value of $\sigma_{FTE,vert} (= 1.5\ m)$, we determine the $VAL_{nominal}$ of Coverages I and II to be 15.3 m and 51.4 m, respectively.

3.2 VALs under the Malfunction NSE Case

Undetected propagation anomalies such as those induced by abnormal ionospheric or tropospheric conditions are considered malfunctions. Errors resulting from malfunctions are assumed as a bias. Therefore, the sum of the biases from the malfunction and nominal error terms must not exceed position uncertainty budgets (50 ft for Coverage I or 150 ft for Coverage II). In the malfunction case, the nominal errors are fixed at their 95th percentiles, as shown in Eq. (4).

$$VAL_{malfunction,Coverage\ I} + TSE_{95\%,vert}(D) \leq 50\ ft, \quad (4)$$
$$VAL_{malfunction,Coverage\ II} + TSE_{95\%,vert}(D) \leq 150\ ft,$$

where D is the distance between the ground facility and UAV. $TSE_{95\%,vert}$ is given by

$$TSE_{95\%,vert} = 1.96\sqrt{\sigma^2_{NSE,vert} + \sigma^2_{FTE,vert}}. \quad (5)$$

In this study, $TSE_{95\%,vert}$ computed using a σ_{FTE} of 1.5 m and σ_{NSE} of 1.7 m. Using Eq. (4) and (5), we determine $VAL_{malfunction}$ for Coverages I and II to be 10.8 m and 41.5 m, respectively. Because the VALs derived from the malfunction case are smaller than $VAL_{nominal}$ in both coverages, we selected $VAL_{malfunction}$ as the final VAL for each region.

4. THEORETICAL PERFORMANCE EVALUATION

In the LAD-GNSS for UAV, two VPLs are computed in real time. The first one is VPL for the nominal case that represents a fault-free condition or faulted conditions. By taking benefits from the known satellite geometry, $VPL_{nominal}$ with optimal allocation of integrity and continuity is computed. The other one is VPL for the malfunction that represents the position error due to an undetected ionospheric anomaly. Since the ionospheric anomaly is a representative malfunction case in navigation system error, the only $VPL_{malfunction}$ in ionospheric anomaly case is computed. The resulting VPL_{final} is determined by taking maximum among those two VPLs. When the VPL_{final} is less than or equal determined VAL in Section 3, UAVs operation in each coverage is available.

4.1 VPL under the Nominal Case

In the current GBAS architecture, separate PL values for fault-free, single reference-receiver fault, and ephemeris-fault cases are defined based on predetermined integrity and continuity allocations. For the remaining fault modes, the ground facility ensures that position uncertainty due to other fault modes not exceed VPL_{H0}. This concept is commonly known as a maximum-allowable error in range (MERR), which causes conservative VPLs that may hold excess integrity allocation. As mentioned in Section 1, by using the known satellite geometry of UAV, the onboard module can compute PLs for all fault scenarios and optimally allocate the integrity and continuity budgets to make the PLs identical for all fault scenarios.

Figure 5 shows the integrity fault tree of a system to which optimal allocation was applied. The total integrity risk requirement of 2×10^{-6} per 150 s is allocated. As UAV accidents can be expected to have fewer casualties than manned aircraft accidents, the relaxed value of GBAS CAT I is allocated. In addition, sub-allocations are adjusted by the inclusion of optimal allocation. In the redesigned integrity fault tree, the integrity risk of single deformation, low signal level, and all other failures are pre-allocated. For the signal deformation integrity risk, a small value (1×10^{-8} per 150 s) is allocated because signal deformation error is canceled out when the ground subsystem and UAV use receivers that are either the same or very similar. 1×10^{-8} per 150 s is also pre-allocated to low signal level integrity risk because the bound of the low signal level is determined based on the performance impact of other monitors, not on the MERR. Finally, as all other failures are bounded by using $VPL_{malfunction}$, the integrity risk—which is the same as that of GBAS CAT I—is allocated.

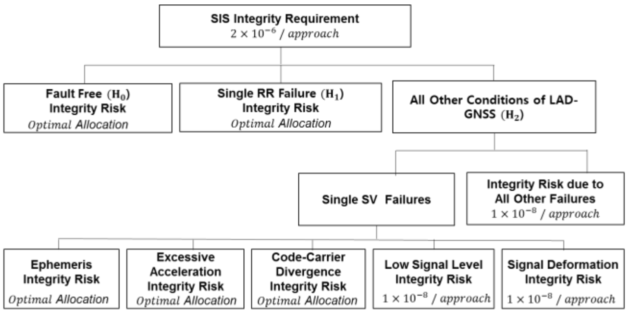

Figure 5. Integrity risk allocation of LAD-GNSS for UAVs

As the allocations are optimized with the PLs for each fault mode, fault mode PLs that were bounded using the MERR should be newly derived. The VPL under the fault hypothesis for satellite i consists of two components, as shown in Eq. (6). First, the position error bound (μ_{fault}) is computed based on bias induced by the satellite fault. Second, the random error term is computed using a sigma-multiplier (k_{HMI}), which considers that monitors miss-detect such faults. As μ_{fault} depends on the fault hypothesis, it is important to find the relationship between the monitor domain errors and user range domain errors to derive the VPLs for the fault mode. In this study, we develop VPLs for excessive acceleration (EA) and code-carrier divergence (CCD).

$$VPL_{fault}(i) = \mu_{fault,i} + k_{HMI,EA}\sum_{i=1}^{n} S_{vert,i}^{2}\,\sigma_i^2, \quad (6)$$

where k_{HMI} is a multiplier defined to ensure a specified probability of the missed detection, and σ_i is the standard deviation of the differential ranging error for satellite i. Additionally, S_{vert} is the row of the weighted-least-squares projection matrix corresponding to the vertical position state [4].

The excessive acceleration threat is a fault condition in a GNSS satellite that causes the range measurements to excessively accelerate. The most probable causes would be GNSS clock anomalies or ground receiver failures by verifying the consistency of the pseudorange and carrier-phase measurements over the last few epochs. The effective differential ranging error ($\delta\rho_{EA}$) due to an acceleration (acc) for satellite i is [5, 6]:

$$\delta\rho_{EA,i} = 0.5\cdot\tau\cdot(\tau+T)\cdot acc, \quad (7)$$

where, latency (τ) represent the time between correction computation from ground subsystem and their application at aircraft side, and T is the measurement update time interval. For the LAD-GNSS ground subsystem, the τ is 1.5 s and T is 0.5 s. Using Eq. (7) and the minimum detectable error (MDE) concept [3], the position error bound (μ_{EA}) for EA is derived as

$$\mu_{EA,i} = S_{vert,i}\cdot 0.5\cdot\tau\cdot(\tau+T)\cdot\left(k_{cont,EA}+k_{HMI,EA}\right)\sigma_{t_{EA}}. \quad (8)$$

Here, k_{cont} is a multiplier defined to ensure a specified probability of a fault-free alarm, and $\sigma_{t_{EA}}$ is the standard deviation of the nominal test statistic variation (acc). Once μ_{EA} is derived, VPL_{EA} can be calculated by adding the random error term to μ_{EA} as shown in Eq. (9).

$$VPL_{EA}(i) = \mu_{EA,i} + k_{HMI,EA}\sum_{i=1}^{n} S_{vert,i}^{2}\,\sigma_i^2. \quad (9)$$

The code-carrier divergence monitor is used to detect ionospheric storms and ensure that the divergence of the code and carrier for any given satellite is sufficiently small. The divergence rate is the estimated change rate of ionosphere delay, I, over time. Given a divergence failure with a CCD rate *Div* and time onset $t=0$, a differential ranging error ($\delta\rho_{CCD,i}$) due to a divergence rate for satellite i is [7]:

$$\delta\rho_{CCD,i} = \left|e_g\left(t-max\left[t_{g,0},0\right]\right)-e_a\left(t-max\left[t_{a,0},0\right]\right)\right|, \quad (10)$$

where e_g is the ranging error due to *Div* at ground relative to ground steady state, and e_a is a ranging error due to *Div* at aircraft relative to ground steady state, $t_{g,0}$, and $t_{a,0}$ is the ground filter start time and airborne filter start time, respectively.

Figure 6 shows the worst-case differential ranging error at *Div* of 0.004 m/s. Note that prior research [7] suggests that a nominal value of σ_{Div} is 0.004 m/s. The worst case occurs when the ground filter has already converged and the aircraft filter has just begun. For a newly acquired satellite, the pseudo-range is used after two smoothing time constant intervals, or 200 s, to reach steady state. Therefore, the worst differential range error at σ_{Div} of 0.004 m/s is 0.074, as shown in Fig. 6. Similar to VPL for EA, the VPL for CCD is derived by using the worst differential range error at σ_{Div} of 0.004 m/s and the MDE concept.

$$VPL_{CCD}(i) = S_{vert,i} \cdot (k_{cont,CCD} + k_{HMI,CCD}) \cdot 0.074 \\ + k_{HMI,CCD} \sum_{i=1}^{n} S_{vert,i}^2 \sigma_i^2. \quad (11)$$

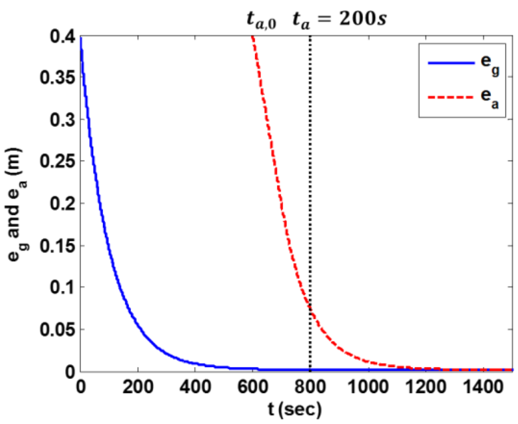

Figure 6. Ranging error due to *Div* of 0.004 m/s relative to the ground subsystem steady state

4.1.1 VPL with Optimal Integrity and Continuity Allocation

The optimal problem for finding the minima of PLs—which are subject to the constraint that the sum of allocations of each fault mode equals the total risk—is solved by using a Lagrangian function [8]. In this method, all VPLs are expressed as a function of the continuity and integrity risk, as shown in Eq. (12):

$$VPL_i\left(P_{cont,i}, \frac{PHMI_i}{P_{ap,i}}\right) = M_i(P_{cont,i}) + L_i\left(\frac{PHMI_i}{P_{ap,i}}\right), (12)$$

where $P_{cont,i}$ and $PHMI_i$ respectively are continuity and integrity allocated to hypothesis i, and $P_{ap,i}$ is the prior probability of hypothesis i. Once the VPLs are organized into the above expression, a two-dimensional optimization problem can be written as

$$\begin{aligned} &\text{Minimize} \quad \max_i \left[M_i(P_{cont,i}) + L_i\left(\frac{PHMI_i}{P_{ap,i}}\right)\right] \\ &\text{subject to} \quad \sum_{i=0}^{N_{mod}-1} P_{cont,i} = P_{cont} \\ &\text{and} \quad \sum_{i=0}^{N_{mod}-1} PHMI_{,i} = PHMI, \end{aligned} \quad (13)$$

where P_{cont} and $PHMI$ are the total continuity and total integrity risk, respectively, and N_{mod} is the number of hypotheses. Using the results of Section 4.1, the VPLs for LAD-GNSS can be expressed as

$$\begin{aligned} VPL_{H0} &= k_{HMI,H0} \sum_{i=1}^{n} S_{vert,i}^2 \sigma_i^2, \\ VPL_{H1}(j) &= k_{HMI,H1} \sum_{i=1}^{n} S_{vert,i}^2 \sigma_{H1,i}^2 + |B_{vert,j}|, \\ VPL_e(i) &= k_{cont,H0} \cdot \left(\frac{|S_{vert,i}||x|}{R_i} \sigma_{t_e}\right) \\ &\quad + k_{HMI,H0}\left(\frac{|S_{vert,i}||x|}{R_i} \sigma_{t_e} + \sum_{i=1}^{n} S_{vert,i}^2 \sigma_i^2\right), \\ VPL_{EA}(i) &= k_{cont,EA,i} \cdot \left(|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T)\sigma_{t_{EA}}\right) \\ &\quad + k_{HMI,H0}\left(|S_{vert,i}| \cdot 0.5 \cdot \tau \cdot (\tau + T)\sigma_{t_{EA}} + \sum_{i=1}^{n} S_{vert,i}^2 \sigma_i^2\right), \\ VPL_{CCD}(i) &= k_{cont,H0} \cdot (|S_{vert,i}| \cdot 0.074) \\ &\quad + k_{HMI,H0} \cdot \left(|S_{vert,i}| \cdot 0.074 + \sum_{i=1}^{n} S_{vert,i}^2 \sigma_i^2\right), \end{aligned} \quad (14)$$

where σ_{H1} is the standard deviation under a single reference-receiver fault, B_{vert} is the B-value converted to a vertical position error, and x is the displacement vector between the reference station and aircraft antennas. In addition, R_i is the distance (scalar) to satellite i. It is practical at this point to change notations and variables so that the notations are simpler. In Eq. (14), we can change the terms multiplied by continuity, the terms multiplied by integrity, and B-value into θ, ψ, and β, respectively. The problem to be solved is then

$$\begin{aligned} &\text{Minimize} \quad \max_i [k_{HMI,i}\psi_i + k_{cont,i}\theta_i + \beta_i] \\ &\text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2a_i Q(-k_{cont,i}) = P_{cont} \\ &\text{and} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI, \end{aligned} \quad (15)$$

where Q is a unit Gaussian cumulative distribution function, a_0 is zero, and $a_i (i \neq 0)$ is one. According to [8], by arbitrarily allocating the continuity budget, the optimal problem can be simplified to a one-dimensional search space. In this suboptimal scheme, parameter $k_{cont,i}$ is therefore no longer a variable, and we can define

$$\beta_i' = k_{cont,i}\theta_i + \beta_i. \quad (16)$$

The problem to be solved is then

$$\begin{aligned} &\text{Minimize} \quad \max_i [k_{HMI,i}\psi_i + \beta_i'] \\ &\text{subject to} \quad \sum_{i=0}^{N_{mod}-1} 2P_{ap,i} Q(-k_{HMI,i}) = PHMI. \end{aligned} \quad (17)$$

In this paper, we implement the suboptimal scheme because the optimal allocation of the continuity offers only a minor improvement while increasing the complexity of the algorithm and computation load [8].

From the suboptimal scheme described above, as well as the VPLs derived in Sections 4.1, simulations of a VPL with an optimal allocation were conducted. For the simulations, the continuity risks were allocated to be the same as those of GBAS CAT-I, and the standard RTCA 24 constellation was used to generate visible satellite geometries. The satellite geometries were simulated every 5 min (which gives 288 epochs for one day).

In addition, we assumed that $\sigma_{t_{EA}}$ is 0.99 cm/s^2 at C/N_0 of 32 dB-HZ and σ_{Div} is 0.004 m/s. For the standard deviation of ground error (σ_{pr_gnd}), the ground accuracy designator (GAD)-A model with M = 1 was used. The latter was used because the effect of mitigating the multipath error is not expected in an environment where the separation distances between the antennas are narrow. The standard deviation of airborne error (σ_{air}) was computed by using the airborne accuracy designator (AAD)-B model for the simulation.

Figure 7 shows the resulting VPLs for Coverage I and II when the UAV operated 5 km apart from the ground subsystem with a constant speed of 15 m/s in Coverage I and 20 km apart from ground subsystem with a constant speed of 45 m/s in Coverage II. In this result, the VPLs for fault-free (H0), single reference-receiver fault (H1), ephemeris fault cases, EA-fault cases, and CCD-fault cases were computed by using the pre-allocated integrity risk (i. e. without optimal allocations).

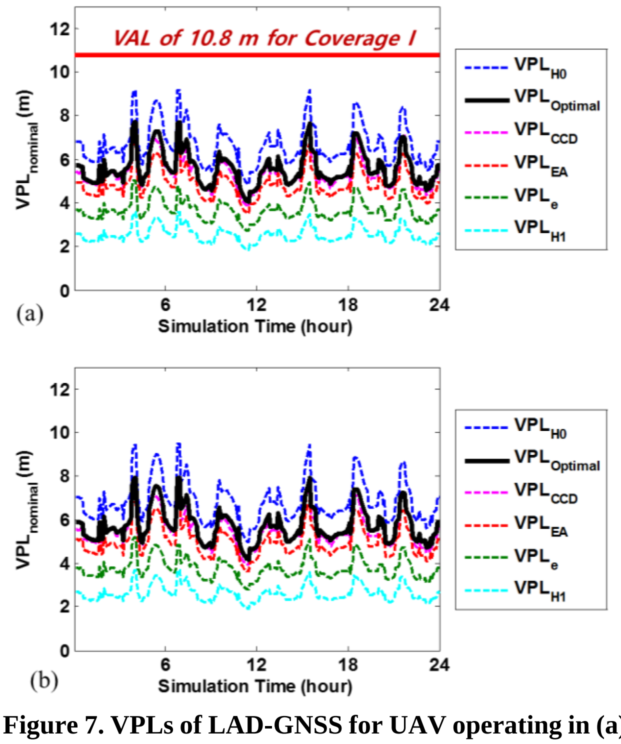

Figure 7. VPLs of LAD-GNSS for UAV operating in (a) Coverage I and (b) Coverage II

As show in Fig. 7, the VPL with the optimal allocation of integrity risk, which is shown by the black curve, is slightly below the VPL_{H0}. These results show that the VPLs of Coverage I and II are reduced by approximately 16% by using the optimal allocation. The VPL with the optimal allocation of integrity is reasonably well-bounded by VAL, which is the operation requirement for Coverage I and II.

4.2 Vertical Protection Level under Malfunction Case

To mitigate the integrity risk due to anomalous ionosphere gradients, several monitors including the ground CCD monitor, DSIGMA, and the simplified geometry screening monitor were implemented in LAD-GNSS. The CCD monitor and DSIGMA could detect and exclude the satellite impacted by large ionospheric gradients using the divergence rate of code-carrier measurements. However, the ionosphere gradients that moved at a slow speed were difficult to detect by both the ground CCD monitor and DSIGMA, which resulted in unacceptable large residual errors in GNSS range signals, even after applying differential corrections. Thus, to assess the impacts of anomalous ionosphere gradients on LAD-GNSS users, the worst differential range error was first tabulated using an exhaustive search over combinations of the simulation parameters in the presence of the ground CCD monitor and DSIGMA. Based on the table of the worst differential range errors, $VPL_{malfunction}$ was computed to evaluate the integrity risk of LAD-GNSS users.

4.2.1 Worst-case Differential Range Error Due to Anomalous Ionosphere Gradient

This section describes how we model the worst differential errors in the range domain caused by anomalous ionosphere gradients. In LAD-GNSS, the code-carrier smoothing is processed to mitigate the receiver noise and multipath on both airborne and ground receivers. However, code-carrier divergence due to ionospheric gradients induces bias in the code-carrier smoothing. Such errors in carrier-smoothed code become larger as either a smoothing time or a magnitude of gradient becomes larger. In LAD-GNSS, both the aircraft and ground facility use a time constant of 30 s for the code-carrier smoothing filter in the presence of the anomalous ionosphere gradient.

The resulting differential range errors can be computed by taking the difference between the smoothed ionospheric delay measurements observed from aircraft and the ground facility. Undetected ionosphere-induced differential range errors are then generated for a hypothetical UAV flight in the presence of the modeled ionospheric gradient. The anomalous ionosphere gradient is modeled with several parameters including the magnitude, width, direction, and relative speed of an ionospheric front with respect to a satellite IPP observed from a ground facility. The moving direction of the UAV and the initial position of both the UAV and ionospheric front are also included in the simulation parameters.

The simulation parameters and ranges are listed in Table 2. For a given set of simulation parameters, the differential range errors and the test statistic of the monitors are computed every epoch as the ionospheric front and UAV advance. When the UAV finally reaches the evaluation point, the differential range error, either detected or undetected, is recorded. Two different UAV operation scenarios for both Coverage I and II are considered in the simulation. The evaluation point is assumed to be 5 and 20 km apart from the ground facility, respectively. The UAV horizontal speed is set at 15 m/s in Coverage I, and 45 m/s in Coverage II, because the speed limit of UAV is constrained to be less than 45 m/s according to the Federal Aviation Administration (FAA) regulations.

Table 2. Simulation parameters

Parameter	Min.	Max.	Step Size
Δv (m/s)	-100	350	10
g (mm/km)	200	500	20
W (km)	25	75	25
α (deg)	-90	90	15
β (deg)	-90	90	10

Figure 8 depicts an example of the results of the worst differential range error in terms of the relative speed of the front. The worst differential range error curves varying with the magnitude of ionosphere gradients from 200 mm/km to 500 mm/km are searched based on the exhaustive simulation over all possible combinations of the ionosphere parameters for each given Δv. The constant model (black line) is proposed to conservatively cover the simulated worst range error curves for all conditions. Since the single-impact scenario is only considered in this analysis, the model of the differential range error does not need to be expressed as a function of Δv. The results of the worst differential range error for the three UAV operation scenarios are summarized in Table 3.

Table 3. Maximum of the worst differential range error for two different UAV operation scenarios

Scenario	Coverage	Horizontal speed of UAV (m/s)	Evaluation Point (km)	Max. range error (m)
(a)	I	15	5	3
(b)	II	45	20	11.5

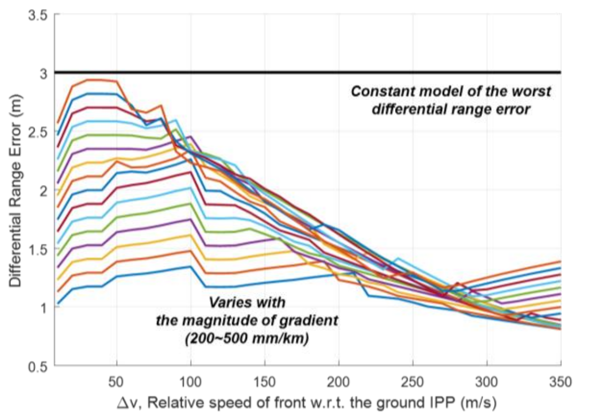

Figure 8. Simulated results of the worst differential range error when the evaluation point is set at 5 km apart from the ground facility and the horizontal speed of UAV is 15 m/s

4.2.2 Vertical Position Error Due to Ionosphere Gradient

The analytical model of the ionosphere-induced differential range error was introduced in Section 4.2.1. Using this model, we examine the largest potential errors resulting from the worst case scenarios. A simulation is conducted to compute the worst-case undetected vertical ionospheric errors (i.e. $VPL_{malfunction}$) by assuming the worst case SV-user geometry and parameter combinations. In this paper, a single-satellite impact is assumed to be the worst case scenario instead of the double-satellite impact mentioned in [9]. This is because the double-impact scenario is a rather conservative assumption for UAV applications.

For a given single satellite *i*, a vertical ionospheric error is defined as:

$$\mathrm{IEV}_i = \left|S_{vert,i}\varepsilon_i\right| \tag{18}$$

where $S_{vert,i}$ and ε_i denote the vertical position component of the weighted-least squares projection matrix S and the ionosphere-induced range error for satellite *i*, respectively. The ionosphere gradient is assumed to move in the worst direction; thus, the resulting Δv is always small. Accordingly, ε_i is always taken as its maximum in the single-impact scenario. Among IEVs computed for all individual satellites in view, the largest IEV is taken as $VPL_{malfunction}$. The final VPL is then determined as the maximum value among $VPL_{malfunction}$ and $VPL_{nominal}$.

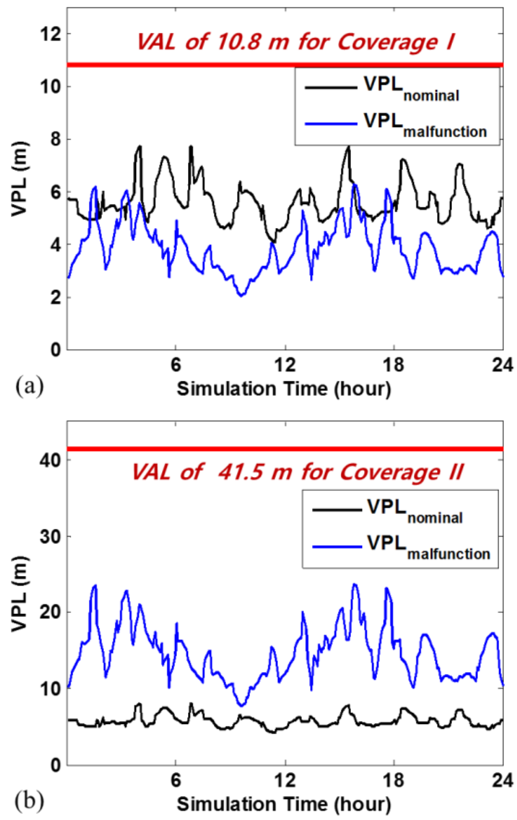

Figure 9. Comparison of the VPL under nominal case, VPL under malfunction case, and VAL for (a) Coverage I and (b) Coverage II

A simulation of $VPL_{malfunction}$ was conducted based on the standard RTCA 24 constellations over the course of a day. For the standard deviation of navigation errors, we used the same model as used in the simulation of VPL under nominal case. $VPL_{malfunction}$ was computed for ideal scenarios. “Ideal” means that the UAV loses no satellite in view. Figure 9(a) shows the result of the $VPL_{malfunction}$ simulation for Coverage I when the UAV operates 5 km apart from the ground facility with a constant speed of 15 m/s. Blue and black curves indicate the $VPL_{malfunction}$ and $VPL_{nominal}$ with the optimal allocation described in Section 4, respectively. The result shows that the final VPLs are determined mainly by $VPL_{nominal}$. On the other hand, in case of Coverage II, the final VPL is determined by $VPL_{malfunction}$ as shown in Fig. 9(b). This is because that the error induced by worst ionospheric gradients sharply increases as the distance between the ground subsystem and the UAV gets larger. As shown in the figure, VPLs for both Coverage I and II were well below the desired alert limits.

5. CONCLUSION

In this paper, we presented an LAD-GNSS architecture for UAVs that communicates with ground subsystems in real time. We fully implemented and analyzed an LAD-GNSS for UAVs, consisting of operation requirement design, integrity monitoring, and theoretical performance evaluation strategies. We determined operation coverages that were defined for two regions with differing operating requirements. For Coverage I, we compared the VPL to a VAL of 10.8 m (determined by analyzing navigation uncertainties in nominal and malfunction conditions). UAV operating Coverage II was expected to meet a VAL of 41.5 m.

In addition, we derived PLs for EA and CCD fault cases to optimally allocate the integrity and continuity budgets. We simulated VPL with optimal allocation; in a simulation of a 24-satellite GPS constellation, VPLs of nominal case were reduced by approximately 16% by implementing optimal allocation.

NSE and FTE errors strongly depended on hardware, environmental, and operational conditions and thus must be experimentally evaluated under various conditions. To this end, our future work will include both simulation-based analysis and field tests of several types of UAVs in different flight environments and modes.

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REFERENCES

[1] S. Pullen, P. Enge, and J. Lee, "High-Integrity Local-Area Differential GNSS Architectures Optimized to Support Unmanned Aerial Vehicles (UAVs)," *Proceedings of ION ITM 2013*, San Diego, CA, Jan. 28-30, 2013.

[2] C. Shively, “Safety Concepts for Mitigation of Ionospheric Anomaly Errors in GBAS,” *Proceedings of ION NTM 2008*, San Diego, CA, January 28-30, 2008.

[3] M. Kim, K. Kim, J. Lee, and S. Pullen, "High Integrity GNSS Navigation and Safe Separation Distance to Support Local-Area UAV Networks," *Proceedings of ION GNSS 2014*, Tampa, FL, September 2014, pp. 869-878.

[4] RTCA SC-159, *Minimum Aviation system Performance Standards for the Local Area Augmentation Systems*, RTCA/DO-245A, Dec. 2004.

[5] G. Xie, “Optimal On-Airport Monitoring of the Integrity of GPS-Based Landing Systems,” Ph.D. Thesis, Stanford University, Stanford, CA, 2004.

[6] M. Brenner and F. Liu "Ranging Source Fault Detection Performance for Cat III GBAS," *Proceeding of ION GNSS*, Portland, OR, 2010.

[7] D. V. Simili, “Code-Carrier Divergence Monitoring for the GPS Local Area Augmentation System,” Master’s Thesis, Illinois Institute of Technology, Chicago, IL, 2007.

[8] J. Blanch, T. Walter, and P. Enge, “RAIM with Optimal Integrity and Continuity Allocations under Multiple Failures,” *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 46, No. 3, 2010.

[9] J. Lee, J. Seo, Y. Park, S. Pullen, and P. Enge, "Ionospheric Threat Mitigation by Geometry Screening in Ground-Based Augmentation Systems," *Journal of Aircraft*, Vol. 48, No. 4 2011, pp. 1422-1433.